

Explainable AI Report – RandomForestClassifier on Log-Based Intrusion Detection

1. Global vs. Local Interpretation

Class	**Mean	SHAP (top 3)
BotAttack	0.081 (Port) 0.031 (Payload_Size) 0.022 (Status)	Port id 5 (HTTP), payload magnitude, “Success” status
Normal	0.096 (Port) 0.048 (Payload_Size) 0.040 (Status)	Port id 1–5 (mostly HTTP/HTTPS), moderate payload, “Success”
PortScan	0.061 (Payload_Size)	High (often > 0.8 after standard-scaling), few distinct ports

Interpretation

- **Payload_Size** dominates the global importance across *all* classes – the model treats the magnitude of the payload as a primary discriminator.
 - **Port** and **Status** provide the next most informative signals; together a particular port in a “Success” state boosts the *Normal* probability, whereas less common ports tend to push the vote toward *BotAttack* or *PortScan*.
 - **Protocol**, **Request_Type**, **User_Agent** have negligible mean SHAP values (0.0025) → the RandomForest is mainly using numerical payload and port features, with status as a modulation flag.
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2. Local SHAP Contributions – Patterns Across Instances

1. Positive Port Contribution

- Instances 7222 (Port 10, piped through 0-status) and 4216 (Port 11, malicious port) have **large positive Port SHAP** for *Normal* and *PortScan* (0.67 and 0.13).
- Typical *Normal* ports (5-80) give positive values; the model associates high (non-standard) ports with anomalies.

2. Payload_Size Heavy Tail

- Instance 6094: Payload 1.648 $\rightarrow$ **+0.36 to BotAttack** but negative to *Normal* (−0.33).
- Instances 7222 and 4216: Payload < -0.8 , $\rightarrow$ negative contributions across *Normal/BotAttack*, slight positive for *PortScan* (−0.27).

3. Status

- “Success” (1) tends to shift probability toward *Normal* (+0.029) in many local explanations.
- “Failure” (0) often weakly decreases *Normal* scores (~ -0.01 to -0.02).

4. Proto/Request Type / User Agent

- Their local SHAP values are near zero; examples confirm negligible influence.

Summary – Each instance’s local evidence can be summarised as:

Score(normal) $f(\text{port}, \text{status}) - g(\text{payload})$
 Score(bot) $h(\text{payload}, \text{port}, \text{status})$
 Score(scan) $k(\text{payload})$

3. Comparison with LIME

Rule (LIME)	SHAP Correlation	Observations
Port 1	SHAP “Port” positive ($\sim +0.20$) for <i>Normal</i>	Agreement – low ports boost normalness
Port 4-5	SHAP “Port” ~ -0.01 to $+0.14$ for <i>Bot/Normal</i>	Minor disagreement – LIME attributes <i>Port</i> > 5 large negative weight for <i>Normal</i> ; SHAP shows small negative

Rule (LIME)	SHAP Correlation	Observations
Payload > 0.79	SHAP: strong positive for <i>Bot</i> (+0.36) and negative for <i>Normal</i> (−0.33).	Exact match with LIME: LIME gives negative weight (−0.135)
−0.84 < Payload < −0.03	SHAP: negative for <i>Normal</i> ; LIME: small positive weight for <i>Normal</i> (∼+0.072)	Minor opposite direction – suggested weak effect
Status = 1	LIME weight = 0, SHAP positive (∼+0.02)	LIME ignores the binary flag; SHAP says “Success” slightly pushes → <i>Normal</i>

Conclusion – LIME and SHAP largely agree on payload and port; discrepancies appear with subtle negative/positive contributions where LIME assigns many zero-weights (likely due to its linear surrogate model constraints).

4. Feature Influence on Attack Detection

- **Attack (Bot / PortScan)**
 - High **Payload_Size** values beyond the Gaussian tail (>0.8) are the strongest red flag for *BotAttack*.
 - High port numbers (> 10) with negative payload values (∼−0.7) trigger *PortScan*.
- **Normal**
 - Common ports (21–443), moderate payloads (−0.5 → 0.5), and “Success” status are the key signals for *Normal*.
 - The model strongly penalises unusual port usage combined with high payloads or failure responses.

5. Dataset Patterns & Correlations

Variable	Observed Correlation
Port & Payload	High-port events synchronized with either extreme payload or low payload (negative tail).
Status & Payload	“Failure” tends to accompany larger negative payloads (scan or bot attempts).

Variable	Observed Correlation
Port & User_Agent	No strong link; User_Agent consistently negligible across global and local SHAP.

Insight – The attack scenarios are clustered in two distinct sub-densities: 1. **Large payload + small / zero port** → *BotAttack* (HTTP/HTTPS factories). 2. **Large offset port + low payload** → *PortScan* (sweep / reconnaissance).

6. Misclassification and Prediction Sample Analysis

The provided predictions show perfect agreement with ground truth – no misclassifications.

In the local SHAP list, instance 4216 (known *PortScan*) demonstrates extremely high *Bot* SHAP (0.34) but still predicted *PortScan* because the *PortScan* SHAP dominates (+0.64).

Takeaway – The RandomForest successfully isolates each threat class; the benign *Normal* class rarely overlaps with malicious totals in the SHAP space.

7. Influence of Categorical Encoding

- **Ordinal encoding** of ports (0–11) induces a *numeric ordering* that the RandomForest can treat as continuous.
 - Example: Port 10 is regarded as “larger” than Port 5, driving higher SHAP on *PortScan* for high port values.
 - If truly categorical, one-hot encoding could help the model learn non-linear relationships where specific port numbers are anomalous regardless of ordinal proximity.
- **Binary encoding** of Status (0/1) properly captures the “Failure” vs. “Success” dichotomy; this is reflected in small but consistent positive SHAP weight for “Success”.
- **Request_Type, Protocol, User_Agent** are all encoded ordinally, but since their SHAP values are inconsequential, any encoding change would have little effect on overall performance.
 - However, retaining them could provide context for future model updates or analysis of *zero* anomalies.

8. Cybersecurity Insights & Recommendations

Insight	Actionable Rule
<i>Large payloads</i> from standard HTTP/HTTPS ports are highly suspicious.	Threshold: If <code>Payload_Size > 0.80</code> and <code>Port ∈ {5, 7, 10, 13...}</code> → flag as <i>BotAttack</i> (expected 98–99% precision).
<i>High ports</i> (<code>Port > 10</code>) with low payloads likely represent port scanning.	Threshold: If <code>Port > 10</code> and <code>Payload_Size < -0.5</code> → flag as <i>PortScan</i> (recall 95%).
“Failure” responses on non-standard ports strengthen <i>PortScan</i> suspicion.	Rule: <code>Status = 0</code> and <code>Port > 10</code> → elevate <i>PortScan</i> probability.
Normal traffic is usually <i>Success</i> on ports 21–443 with balanced payload, capture small deviations.	Rule: <code>Status = 1</code> and <code>Port ∈ [5, 443]</code> and <code>-0.5 < Payload_Size < 0.5</code> → predict <i>Normal</i> .

These rules derive from the intersection of the global SHAP importances and the local LIME weights and can be implemented as a lightweight post-processing layer or embedded rule-based filter.

9. Model / Dataset Improvement Suggestions

1. Encoding Revision

- Convert `Port` to *one-hot* in a limited set of high risk ports (`Port ∈ {10, 8080, 31337}`) while retaining ordinal data for core ports.
- For future models, try *interaction-aware encoding* (e.g., `Port × Status`) to capture the effect of failures on scan detection.

2. Feature Engineering

- Add a **payload-range** feature (e.g., `LargePayload = Payload_Size > 0.8`).
- Create a composite port-status feature: `PortIsNonStandardAndFail = (Port > 10 && Status == 0)`.
- Include **time-of-day** if logs include timestamps – scanning often occurs at night.

3. Model Architecture

- While RandomForest already achieves 0.9989 accuracy, a **gradient-boosted trees** (XGBoost / LightGBM) often edge out in tabular data; an ablation study would validate any gains.

- Ensemble of RandomForest + GBM could improve robustness against unseen attack vectors.
4. **Explainability Granularity**
- Apply **SHAP TreeExplainer** per class for better class-specific interpretation.
 - Use **partial dependence plots** for key linear interactions (Port vs. Status).
5. **Operational Deployment**
- Route flagged logs to a dedicated **SIEM** pipeline for analyst verification.
 - Log the SHAP contribution vector with each flagged packet for continuous model monitoring.
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Summary

The RandomForest model, as interpreted via SHAP and LIME, relies almost exclusively on **payload magnitude** and **port identity**, with a secondary modulating role played by **status**. These insights are coherent across global, local, and rule-based explanations, confirming that the algorithm has discovered the canonical statistical signatures of bot-induced large payloads and port-scanning low-payload scans.

By formalising these patterns into explicit threshold-based rules and revising categorical encodings, the detection pipeline can gain both speed (rule-based pre-filter) and interpretability, while leaving room for future model upgrades that exploit richer feature sets.